

The Bathroom as a VR Space: How Visual-Tactile Congruence Shapes Presence and Naturalness during Bathing

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ABSTRACT

Japanese bathrooms hold the potential to become novel VR spaces. However, bathrooms remain unexplored as VR spaces, and almost no research has been conducted. To explore this potential, we created a multisensory system for use in a real bathroom and conducted the first exploratory user study on VR experiences during full-body immersion in hot water. Our results demonstrated that visual-tactile congruence significantly impacted user perception (presence and naturalness). These findings highlight the critical role of visual-tactile congruence in modulating the bathing experience and suggest that the bathroom holds potential as a novel space for VR applications.

Index Terms: bathing, visual-tactile congruence, virtual reality, presence, naturalness.

1 INTRODUCTION

Japanese bathrooms typically integrate the bathtub and washing area into a single enclosed space. This self-contained, private environment is well-suited for delivering controlled, immersive sensory experiences. As a result, these bathrooms are ideal spaces for VR experiences. Although there are examples of VR systems designed for aquatic environments, such as AquaCAVE [8] and GroundFlow [1], these systems require highly specialized equipment, making them impractical for use in standard household bathrooms.

We selected a bathroom equipped with a circulating bath water (CBW) system as the experimental setting. The CBW system circulates hot water throughout the bathtub, allowing users to experience water flowing over their neck and shoulders [4]. The sensation of water flow provided by the CBW system may also have the potential to evoke a feeling of being immersed in nature, such as in a river.

In cognitive psychology, it has been suggested that people intuitively value natural environments, making the concept of "naturalness" important in shaping the motivation and evaluation of experiences [6]. Perceptions of naturalness are influenced by both unimodal and multimodal sensory input [7]. Therefore, providing visual and tactile stimuli during bathing may transform the otherwise sterile bathroom setting into an environment that evokes a heightened sense of naturalness.

It has been stated that multisensory VR experiences enhance the user's presence. Among these, tactile stimulation is the most commonly used [3]. Tactile stimuli that are congruent with visual stimuli have been reported to increase the presence [2]. Therefore, designing visual-tactile congruence in multisensory VR experiences during bathing has the potential to enhance presence.

In this study, we attached a water vibration device directly to the bathtub, enabling the delivery of tactile stimuli via water vibrations during bathing. The tactile sensations produced by water vibrations can be modulated by varying their frequency [5], allowing us to

present stimuli that are either congruent or incongruent with the accompanying visual information.

The main objective of this study was to investigate how visual-tactile congruence shapes presence and naturalness during bathing. Fig. 1 shows a participant actually experiencing VR in the bathroom.

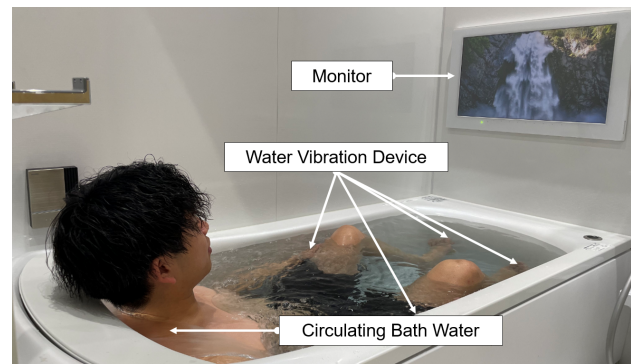


Figure 1: A participant bathing in the created bathroom.

2 METHOD

This experiment was approved by the Ethics Committee for Human Subject Research at the University of Electro-Communications (H24076).

In this experiment, two types of tactile stimuli and two types of visual stimuli were prepared. Regarding the tactile stimuli, prior research indicates that low-frequency water vibrations evoke a "dynamic" impression, while high-frequency vibrations evoke a "static" one [5]. Based on this, we extracted pink noise in two frequency bands to present as Dynamic Haptics (DH: 45-180 Hz) and Static Haptics (SH: 180-710 Hz). For the visual stimuli, two types of natural landscape videos were prepared—a Dynamic Scene (DS: waterfall) and a Static Scene (SS: stream)—as shown in Fig. 2. By combining these, we established four multisensory conditions (I: DH × DS, II: SH × DS, III: SH × SS, and IV: DH × SS).

In this study, a special experimental environment and equipment were utilized to evaluate the multisensory experience during bathing, with its system diagram shown in Fig. 3. The experiment was conducted in a bathroom equipped with a unit bath¹. The flow rate of the CBW was set to 65 L/min. The bath water temperature was controlled within the thermoneutral zone (approximately 37°C) to ensure participants focused on visual and tactile stimuli. The water vibration devices consisted of vibration speakers². The voltage applied to the vibration speakers was standardized at 1.5V. A 24-inch monitor was placed in front of the user. The frequency of the water vibrations was controlled using an equalizer³. The voltage applied to the vibration speakers was amplified using an amplifier⁴.

¹"SYNLA" by TOTO Ltd.

²"656030" by Foster Electric Company Ltd.

³"DCX2496-ULTRADRIVE PRO" by Behringer

⁴"POWER AMPLIFIER PA2030A" by Yamaha Co.

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A total of 30 male volunteers (age $M = 37.2, SD = 4.1$) were recruited as participants. They experienced all four multisensory conditions with the CBW continuously operating. After experiencing each of these multisensory conditions, participants answered the subjective evaluation items (presence and naturalness). Naturalness was evaluated using a 7-point Semantic Differential (SD) method (very natural: 3, natural: 2, somewhat natural: 1, neither: 0, very artificial: -3, artificial: -2, somewhat artificial: -1). Presence was evaluated using a 5-point Likert scale (None at all: 1, Slightly: 2, Moderately: 3, Quite: 4, Very much so: 5).

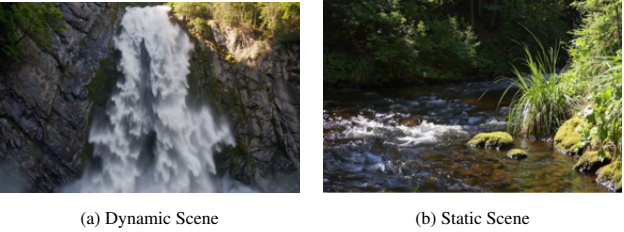


Figure 2: Videos prepared as visual stimuli.

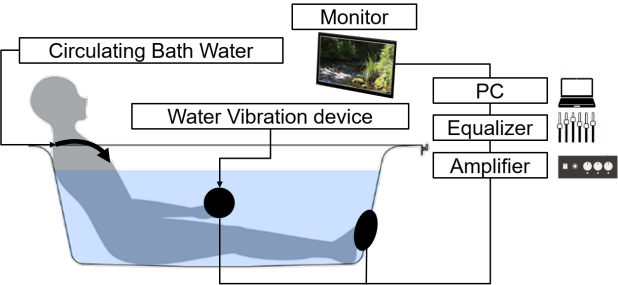


Figure 3: System diagram of the VR experience bathroom.

3 RESULTS & DISCUSSION

Data from all 30 participants were used in the analysis. All statistical analyses were performed using R version 4.3.1.

Regarding the subjective evaluation results, 2×2 analysis of variance was performed. For Presence, the analysis revealed a significant interaction effect ($F(1, 116) = 9.87, p < 0.01$). Significant main effects were also found for Factor tactile ($F(1, 116) = 5.55, p < 0.01$), and Factor visual ($F(1, 116) = 8.67, p < 0.05$). For naturalness, the analysis also revealed a significant interaction effect ($F(1, 116) = 12.28, p < 0.01$). A significant main effect was found for Factor tactile ($F(1, 116) = 7.52, p < 0.01$), but not for Factor visual ($F(1, 116) = 2.32, p > 0.05$). A post-hoc Bonferroni test was conducted to confirm significant differences between conditions. The results are shown in Fig. 4. These results, indicating a significant visual-tactile interaction effect on presence and naturalness, suggest that visual-tactile sensory integration occurred during the bathing experience, potentially altering user cognition and perception.

For dynamic scenes, the congruent condition resulted in significantly higher presence ratings compared to the incongruent condition. However, visual-tactile congruence did not significantly influence naturalness. For static scenes, conversely, congruence did not significantly affect presence. In contrast, the incongruent condition led to a significant decrease in naturalness compared to the congruent condition. These asymmetrical results are likely attributable to the varying degrees of visual-tactile congruence between conditions. Although two types of simplified tactile stimuli were prepared for this study, they may not have adequately reproduced the tactile sensations corresponding to each scene. As a result, the congruence in

the static scene may have been weaker than in the dynamic scene. This could explain why the weaker congruence in the static scene was insufficient to boost presence, but still strong enough to maintain a sense of naturalness that is lost under incongruent stimuli. Since the degree of congruence between sensory stimuli was not verified in this study, examining this in future research will enable a deeper discussion.

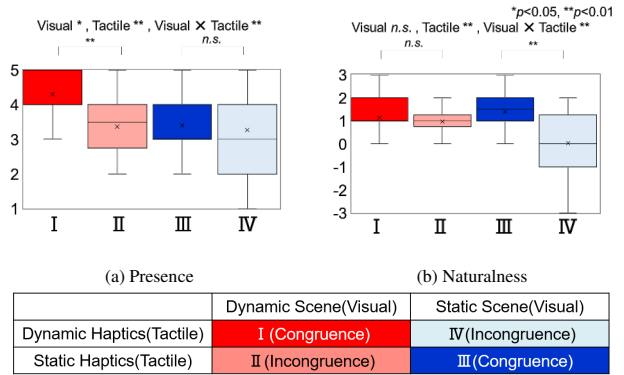


Figure 4: Summary of subjective evaluation.

4 CONCLUSION

To our knowledge, this is the first study to investigate VR experiences during bathing. We developed a bathroom environment capable of delivering VR experiences and conducted a user study to evaluate its effects. This work lays the foundation for future explorations into the diverse impacts of VR experiences in the bathroom on human perception.

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